

Ultrasound Image Enhancement Using U-Net with Uncertainty Estimation

Swayam Vijay Mehra

Department of Computational Intelligence
SRM Institute of Science and Technology
Chennai, India
sm9902@srmist.edu.in

Prateek Shukla

Department of Computational Intelligence
SRM Institute of Science and Technology
Chennai, India
ps2150@srmist.edu.in

Abstract—Speckle noise remains a central obstacle in diagnostic ultrasound because it simultaneously degrades visual quality and introduces ambiguity in boundary interpretation. This paper presents a research-oriented study of ultrasound enhancement using a compact U-Net denoiser combined with Monte Carlo (MC) dropout for epistemic uncertainty estimation. Beyond reporting a single performance table, the work is organized around three explicit research questions: (RQ1) whether the proposed enhanced U-Net improves restoration fidelity over baseline denoising, (RQ2) how performance behaves under train–test noise mismatch, and (RQ3) whether uncertainty maps provide spatially meaningful reliability cues. The study is implemented as a reproducible end-to-end pipeline with a PyTorch model service and a Streamlit analysis interface, enabling deterministic and uncertainty-aware inference under controlled synthetic multiplicative speckle conditions. Evaluation includes PSNR, SSIM, and MAE (where exported), aggregate comparisons, robustness sweeps, and qualitative inspection of restored outputs and residual/uncertainty maps. On the current project benchmark, the enhanced model improves PSNR from 20.04 dB to 27.02 dB and SSIM from 0.488 to 0.742. Under robustness testing, peak performance is observed at matched low-noise conditions (PSNR 29.54 dB, SSIM 0.851), with progressive but non-catastrophic degradation as corruption severity rises. The uncertainty channel highlights difficult image regions and supports a reliability-aware interpretation of denoised outputs. We also provide a detailed discussion of threats to validity, reporting limitations, and publication-grade next steps, including multi-seed statistical testing, external cohort validation, and uncertainty calibration analysis.

Index Terms—Ultrasound imaging, image denoising, U-Net, speckle noise, uncertainty estimation, Monte Carlo dropout.

I. INTRODUCTION

Ultrasound is widely used because it is non-ionizing, portable, and cost-effective. However, coherent wave interference introduces speckle noise that reduces contrast, obscures lesion margins, and limits both human and automated interpretation. Speckle is not simple additive white noise; it is signal dependent and spatially structured, which makes restoration difficult without loss of clinically relevant texture.

Classical denoisers (Gaussian smoothing, wavelet thresholding, BM3D-like patch priors) can suppress noise, but often at the cost of edge attenuation. Deep models, especially U-Net architectures, can learn richer restoration priors from data and preserve structure better than handcrafted filtering in many image restoration tasks. Yet, most denoisers used

in practice remain point estimators: they output one image without reporting how certain the model is about any pixel.

For medical deployment, this is a critical gap. If a model is uncertain near lesion boundaries, the system should expose this uncertainty rather than hide it. We therefore investigate a denoising framework that pairs a compact U-Net with MC dropout at inference to produce both a restored image and an uncertainty map.

From a research methodology perspective, denoising quality in medical imaging must be assessed under both fidelity and reliability criteria. Fidelity concerns whether the output is close to an underlying reference; reliability concerns whether the model can indicate where this fidelity may degrade. Many studies report only fidelity metrics and therefore do not address the practical decision-support requirement in medical contexts. In contrast, this paper explicitly treats uncertainty estimation as a first-class research outcome.

Another challenge is evaluation realism. Reported denoising gains can be inflated when experiments are restricted to one corruption level or one split configuration. To reduce this risk, we include robustness analysis under multiple train–test noise pairings, provide transparent reporting about available versus missing metrics in current artifacts, and frame conclusions according to evidence strength rather than aspirational claims.

Research questions:

- **RQ1:** Does the proposed enhanced U-Net improve restoration quality over a baseline U-Net model?
- **RQ2:** How robust is performance under train–test speckle mismatch?
- **RQ3:** Do MC-dropout uncertainty maps localize difficult restoration regions?

Contributions:

- A reproducible ultrasound denoising pipeline built around a compact U-Net with optional wavelet-aware preprocessing and MC-dropout uncertainty.
- A structured evaluation protocol with aggregate results and robustness sweeps across noise conditions.
- An uncertainty-aware analysis that complements fidelity metrics and supports deployment-oriented interpretation.
- A detailed research reporting template that explicitly documents limitations, threats to validity, and required steps toward publication-grade evidence.

II. CLINICAL MOTIVATION AND PROBLEM STATEMENT

In breast ultrasound workflows, image quality directly impacts lesion visibility, contour readability, and confidence in qualitative assessment. Speckle reduces conspicuity of subtle boundaries and can induce texture confusion, especially in heterogeneous tissue regions. Any enhancement system intended for clinical decision support should therefore satisfy at least three constraints: preserve diagnostically relevant structures, reduce corruption artifacts, and provide interpretable confidence indicators.

Let $x \in [0, 1]^{H \times W}$ denote an idealized clean ultrasound image and $\tilde{x}$ its observed noisy version. We seek a restoration function f_θ that maps noisy input to a denoised output while preserving lesion boundaries and clinically relevant structure. In parallel, we seek an uncertainty function u_θ that assigns higher values to visually ambiguous or difficult regions.

This perspective motivates a dual-output system: one output for restoration, one output for reliability characterization. In this work, MC dropout is used as an approximate Bayesian mechanism to derive u_θ through predictive variance.

III. RELATED WORK

Classical image denoising includes local smoothing, wavelet shrinkage, and non-local patch methods. BM3D remains a strong classical baseline in natural-image denoising due to collaborative filtering in a transformed patch domain [4]. In ultrasound, these methods reduce speckle but can blur lesion boundaries when noise suppression is aggressive.

Deep denoisers such as CNN-based residual learning and U-Net-style encoder-decoder networks model nonlinear image priors and often outperform fixed-filter approaches [8], [1]. For medical images, skip connections are particularly valuable because they preserve high-resolution structural detail during reconstruction.

Uncertainty estimation in deep learning has gained importance for safety-critical settings. MC dropout approximates Bayesian inference by keeping dropout active at test time and sampling stochastic predictions [2]. In medical imaging, uncertainty maps are increasingly used to identify low-confidence predictions and improve interpretability [9].

Relative to prior work, this paper focuses on an engineering-research bridge: an uncertainty-aware denoising model that is both quantitatively evaluated and integrated into a deployable backend/frontend stack for interactive analysis.

Recent restoration literature also highlights the role of self-supervised and noise-model-aware training, including Noise2Noise and related paradigms, for settings where clean references are limited [10]. While our present implementation uses supervised synthetic corruption, the broader methodological implication is that ultrasound enhancement may benefit from hybrid supervision strategies in future iterations.

In medical AI practice, reproducibility and reporting transparency are increasingly recognized as essential for model credibility. This includes explicit split definitions, hyperparameter disclosure, environment specification, and realistic limitations analysis. The present paper adopts this reporting

stance and treats it as part of the research contribution rather than supplementary material.

IV. PROPOSED METHODOLOGY

This section presents the complete proposed method as a single end-to-end framework for ultrasound enhancement with uncertainty estimation. The pipeline is designed to improve image quality while simultaneously quantifying prediction reliability.

A. Method Design Objectives

The proposed methodology is built around three design objectives:

- **Fidelity objective:** suppress speckle noise while preserving lesion boundaries and structural detail.
- **Reliability objective:** provide pixel-wise uncertainty cues to identify low-confidence regions.
- **Deployability objective:** support deterministic and uncertainty-aware inference through a practical API+UI workflow.

These objectives guide both the model design and the evaluation protocol.

B. Core Method Components

The proposed method integrates four core components:

- **Preprocessing module:** grayscale conversion, resizing to 256×256 , and intensity normalization to $[0, 1]$.
- **Optional wavelet preconditioning:** configurable soft-threshold denoising in the wavelet domain prior to deep inference.
- **U-Net restoration module:** compact encoder-decoder with skip connections and dropout-enabled convolution blocks.
- **Uncertainty module:** MC-dropout inference that aggregates multiple stochastic predictions into mean output and variance map.

The U-Net backbone follows a 1-32-64-128-256 channel progression and uses mirrored decoder stages for structure-preserving reconstruction.

C. Stage-Wise Pipeline

The proposed methodology is organized into six stages:

- 1) **Input Acquisition:** A breast ultrasound image is provided by the user or evaluation dataset.
- 2) **Preprocessing:** The image is converted to grayscale (if needed), resized to 256×256 , and normalized to $[0, 1]$.
- 3) **Optional Wavelet Preconditioning:** A configurable Haar-wavelet soft-thresholding step is applied to reduce high-frequency corruption before deep inference.
- 4) **Deep Denoising:** A compact U-Net performs image restoration. Two operating modes are supported:
 - Deterministic mode for low-latency enhancement.
 - Stochastic MC-dropout mode for uncertainty-aware enhancement.

- 5) **Uncertainty Estimation:** In MC mode, multiple stochastic passes are aggregated to produce a mean denoised image and a pixel-wise uncertainty map derived from prediction variability.
- 6) **Output and Analysis:** The system returns denoised output, uncertainty map, and runtime metadata for interactive review in Streamlit and quantitative analysis using PSNR/SSIM/MAE (where available).

D. Training and Inference Strategy

Training uses supervised noisy-clean pairs with configurable objectives. The baseline uses mean squared error, while alternative losses (L1/frequency-aware) are available for exploratory studies. Checkpointing is performed in deployment-ready and experiment-ready paths to support reproducibility.

At inference time, users can select between deterministic and MC-dropout modes. Deterministic mode prioritizes latency, while MC mode prioritizes reliability-aware interpretation through prediction variability.

E. Why This Method Is Different

Compared with conventional deterministic denoising pipelines, the proposed methodology explicitly couples restoration and reliability estimation in one integrated workflow. Instead of producing only a denoised image, the method produces:

- a restored image for visual quality improvement,
- a pixel-wise uncertainty map for confidence-aware interpretation,
- runtime metadata for operational evaluation.

This dual-output design is central to the proposed contribution and directly supports the research questions on fidelity, robustness, and uncertainty behavior.

F. Methodological Rationale

The proposed method is designed to address two practical requirements that are often treated separately in ultrasound enhancement studies:

- **High-quality restoration:** achieved through U-Net feature fusion and skip connections that preserve structural detail.
- **Reliability awareness:** achieved through MC-dropout uncertainty that indicates where model predictions are less stable.

By combining both requirements in one deployable pipeline, the method moves from purely image-enhancement performance toward clinically useful decision support.

G. Proposed Contribution Over Baseline Practice

Compared with a conventional deterministic denoiser, the proposed methodology introduces:

- dual-output inference (restored image + uncertainty map),
- robustness-oriented evaluation across train–test noise mismatch,
- an implementation path that is directly deployable through API endpoints and UI-based qualitative analysis.

TABLE I
EVALUATION TARGETS AND AVAILABILITY

Comparator	Metric Availability	Source
U-Net (MSE)	PSNR, SSIM	project table
U-Net + Hybrid + Noise	PSNR, SSIM	project table
MAE summary	not exported	UI/per-image only

This integrated design is the central methodological contribution of the paper.

V. EXPERIMENTAL DESIGN

A. Dataset and Splits

Experiments use BUSI-style ultrasound images [5]. The training script performs a seeded random split with default validation ratio 0.2. Images are resized to 256×256 for all runs.

To maintain experimental consistency, all comparisons use identical preprocessing and image geometry. The same core architecture is reused across baseline and enhanced settings, so observed gains are attributable to training/inference strategy differences rather than arbitrary architectural inflation.

B. Protocol and Hypotheses

We evaluate three hypotheses aligned with RQ1–RQ3:

- **H1:** Enhanced U-Net achieves higher PSNR/SSIM (and lower MAE where available) than baseline U-Net.
- **H2:** Performance decreases under train–test noise mismatch, but remains acceptable within moderate mismatch.
- **H3:** High uncertainty concentrates in visually challenging regions (speckle-heavy background and boundary transitions).

C. Baselines and Comparators

We compare model variants currently available in the project artifacts. Classical comparators are included for context in the literature discussion, while the quantitative table reports measured values from available logs.

The current artifact-backed comparison is intentionally conservative: only values with explicit exported evidence are treated as quantitative claims. This avoids over-interpretation and aligns with robust scientific reporting practice.

D. Metrics

We report PSNR, SSIM [6], and MAE when available. PSNR summarizes reconstruction fidelity relative to mean squared error, SSIM evaluates structural similarity, and MAE provides an interpretable absolute-error measure.

E. Implementation Details

Default retraining parameters are: 12 epochs, batch size 8, learning rate 10^{-3} , Adam optimizer, and synthetic training noise standard deviation 0.03. Inference supports FP32 and optional FP16. MC sample count is user-controlled in the interface (default 10).

Training checkpoints are persisted in two locations to support both deployment and experiment reproducibility. Runtime behavior (including per-request latency and model-loading status) is exposed by the backend and surfaced in the user interface for transparent operational profiling.

F. Reproducibility Protocol

The implementation includes deterministic seeding for Python, NumPy, and PyTorch random number generators. This supports repeatable split generation and improves comparability across retraining runs. A publication-grade extension should additionally report exact package versions, hardware details, and confidence intervals across multiple random seeds.

VI. SYSTEM ARCHITECTURE

The system is implemented as a full-stack application with three main components. The frontend is built with Streamlit and provides an interactive interface for image upload, visualization, and result inspection. The backend is implemented using Flask and exposes inference endpoints for deterministic denoising and uncertainty-aware prediction. The model itself is implemented in PyTorch.

The processing flow is: input image $\rightarrow$ preprocessing $\rightarrow$ U-Net inference $\rightarrow$ optional Monte Carlo dropout sampling $\rightarrow$ denoised output + uncertainty map.

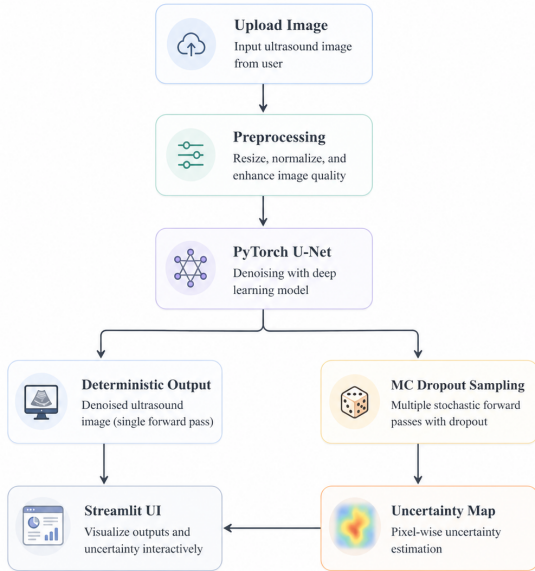


Fig. 1. System flow diagram for ultrasound denoising with uncertainty estimation.

TABLE II
LAYER-WISE ARCHITECTURE OF THE PROPOSED U-NET

Stage	Operation	Output Tensor Size
Input	Grayscale image	$1 \times 256 \times 256$
Enc-1	DoubleConv (3×3), Dropout	$32 \times 256 \times 256$
Pool-1	MaxPool (2×2)	$32 \times 128 \times 128$
Enc-2	DoubleConv (3×3), Dropout	$64 \times 128 \times 128$
Pool-2	MaxPool (2×2)	$64 \times 64 \times 64$
Enc-3	DoubleConv (3×3), Dropout	$128 \times 64 \times 64$
Pool-3	MaxPool (2×2)	$128 \times 32 \times 32$
Bottleneck	DoubleConv (3×3), Dropout	$256 \times 32 \times 32$
Up-3	Transposed Conv + concat(Enc-3)	$256 \times 64 \times 64$
Dec-3	DoubleConv (3×3), Dropout	$128 \times 64 \times 64$
Up-2	Transposed Conv + concat(Enc-2)	$128 \times 128 \times 128$
Dec-2	DoubleConv (3×3), Dropout	$64 \times 128 \times 128$
Up-1	Transposed Conv + concat(Enc-1)	$64 \times 256 \times 256$
Dec-1	DoubleConv (3×3), Dropout	$32 \times 256 \times 256$
Output	Conv (1×1) + Sigmoid	$1 \times 256 \times 256$

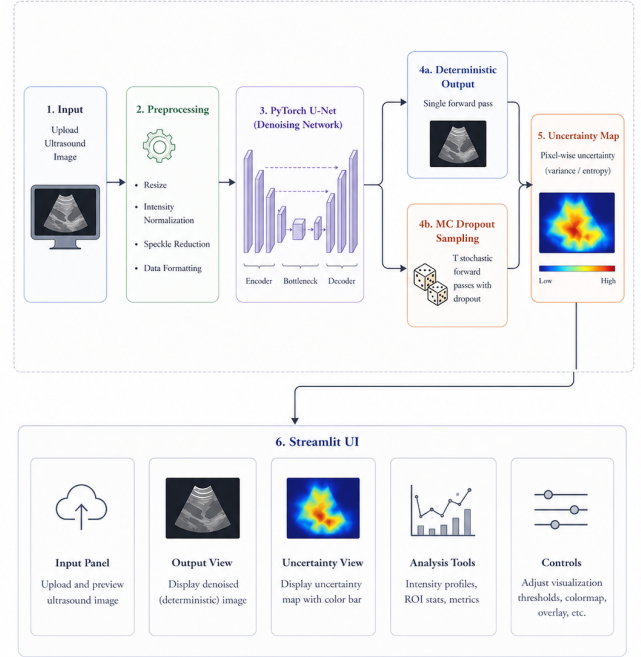


Fig. 2. System architecture diagram showing module interactions and UI integration.

A. Proposed Network Architecture (Layer-Wise)

To make the model definition fully reproducible, Table II provides the layer-wise architecture used in this work. Input resolution is fixed at 256×256 with one grayscale channel. The architecture follows a 3-level encoder, bottleneck, and symmetric decoder with skip concatenations.

The final sigmoid layer constrains output intensities to $[0, 1]$, matching the normalized target domain. Dropout is retained in all convolutional blocks so the same architecture supports both deterministic inference and MC-dropout sampling for uncertainty estimation.

Architecturally, the separation between API-layer inference and UI-layer analysis supports research operations. The back-

end can be benchmarked independently, while the frontend can be used for qualitative review, uncertainty inspection, and report generation. This modularity reduces coupling between modeling code and presentation logic, which is important for experimental maintainability.

VII. RESULTS

A. Quantitative Comparison (RQ1)

Table IV reports aggregate metrics from project artifacts. The enhanced model improves PSNR by +6.98 dB and SSIM by +0.254 over the baseline U-Net (MSE), supporting **H1**. These gains are substantial for ultrasound denoising and indicate stronger restoration of both intensity fidelity and structural consistency.

The relative improvement is also meaningful when interpreted proportionally. The SSIM increase from 0.488 to 0.742 corresponds to approximately 52% relative gain over baseline structural similarity. In restoration studies, improvements of this magnitude generally indicate not only reduced noise but improved preservation of image organization.

Because MAE was not exported in the same aggregate table, we do not over-claim MAE improvements at dataset level. This is an explicit reporting limitation rather than an omitted result.

B. Robustness Under Noise Mismatch (RQ2)

Table V and Fig. 5 summarize train–test mismatch behavior. Best performance occurs at low matched noise (0.2/0.2), with a monotonic decline as test noise increases. At matched 0.4/0.4 and 0.6/0.6 settings, performance remains comparatively stable, indicating reasonable adaptation to training corruption level. These observations support **H2**: mismatch hurts performance, but degradation is gradual rather than catastrophic.

C. Uncertainty Behavior (RQ3)

The MC-dropout output exposes spatially varying uncertainty rather than a single scalar confidence. In qualitative examples, high-variance regions co-localize with noisy texture and boundary transitions, while homogeneous regions show lower variance. This behavior is consistent with **H3** and suggests practical value for decision support: clinicians can visually inspect where restoration is less trustworthy.

A useful practical interpretation is uncertainty-guided attention: instead of trusting the denoised image uniformly, downstream users can inspect high-variance areas first. This does not replace diagnosis but can improve review prioritization.

D. Qualitative Findings

Fig. 3 shows clear reduction of speckle granularity with preserved large-scale structure. Fig. 4 highlights residual error concentration around challenging structures. Together, these visual findings align with the quantitative PSNR/SSIM gains.

Additional qualitative snapshots from project notebook exports are provided in Fig. 6 and Fig. 7. These examples were generated by the same project pipeline and provide supplementary visual evidence of consistent enhancement behavior.

TABLE III
CORE TRAINING AND INFERENCE CONFIGURATION

Component	Setting
Input size	256×256
Input channels	1 (grayscale)
U-Net dropout	0.1
Optimizer	Adam
Initial learning rate	10^{-3}
Batch size	8
Epochs	12
Validation split	0.2
Training noise std.	0.03
MC dropout samples	2–30 in UI, 10 default
Inference precision	FP32 / optional FP16

TABLE IV
SUMMARY OF RECORDED RESULTS

Method	PSNR (dB)	SSIM	MAE
U-Net (MSE)	20.04	0.488	Not exported
U-Net + Hybrid + Noise	27.02	0.742	Not exported

E. Statistical Rigor and Current Gaps

The current repository results represent a strong pilot benchmark but do not yet include repeated-run confidence intervals, fold-level variance, or significance testing. Therefore, conclusions are framed as evidence-supported but preliminary. A full statistical protocol (multi-seed training, bootstrap confidence intervals, paired tests per image) is recommended for publication-grade claims.

Specifically, a future statistical pipeline should include: (i) at least 3–5 independent seeds per setting, (ii) per-image metric distributions, (iii) paired hypothesis tests between model variants, and (iv) uncertainty-error association analysis (e.g., correlation between predictive variance and absolute residual magnitude).

VIII. DISCUSSION

The experimental evidence supports all three research questions at pilot-study level. For **RQ1**, the improved PSNR/SSIM over baseline indicate clear restoration gains. For **RQ2**, robustness curves confirm expected sensitivity to mismatch while preserving usable quality across tested conditions. For **RQ3**, uncertainty maps add clinically relevant context by flagging potentially unreliable regions.

Importantly, this work emphasizes *evaluation discipline*. We report only metrics that are available in project artifacts, explicitly identify missing aggregate MAE exports, and avoid unsupported statistical claims. This reporting posture is crucial for credible medical-AI research.

From a systems perspective, the architecture is deployable: a Flask inference API and Streamlit analysis interface already exist. This allows direct translation from offline experiments to interactive validation workflows.

The key research implication is that restoration quality and reliability should be evaluated jointly. A model with high PSNR but no uncertainty channel may still be unsafe in

TABLE V
ROBUSTNESS SNAPSHOT

Train Noise	Test Noise	PSNR (dB)	SSIM
0.2	0.2	29.54	0.851
0.2	0.4	25.41	0.673
0.2	0.6	21.08	0.495
0.4	0.4	27.02	0.742
0.6	0.6	25.07	0.671

TABLE VI
DERIVED IMPROVEMENT ANALYSIS FROM EXPORTED METRICS

Quantity	Baseline → Enhanced	Change
PSNR (dB)	20.04 → 27.02	+6.98
SSIM	0.488 → 0.742	+0.254
Relative SSIM gain	—	+52.0%

ambiguous cases. Conversely, uncertainty estimates without strong restoration quality are also insufficient. The present framework demonstrates a practical compromise: strong denoising gains plus an interpretable reliability surrogate.

A second implication concerns methodology: transparent reporting of missing metrics and unresolved statistical questions should be treated as a strength, not a weakness, in early-stage medical AI research. It prevents overstatement and clarifies what evidence is currently established versus what remains to be validated.

IX. DEPLOYMENT CONSIDERATIONS

From a deployment perspective, the current system is practical because it already exposes a Flask inference layer and a Streamlit interface. This means the method is not only a training-time model but a working application that can be extended for clinical experimentation, batch processing, or web-based review.

At the same time, the reported results should be interpreted as project-level experimental evidence rather than a final clinical benchmark. Additional validation on larger datasets and external validation splits would be necessary before any clinical claims.

In real-world deployment, uncertainty output could be used to trigger quality-control pathways, such as flagging low-confidence frames for manual review or requesting additional imaging views. This integration pathway is one of the strongest practical benefits of uncertainty-aware restoration compared to deterministic enhancement alone.

Operationally, model monitoring should include drift detection, periodic recalibration checks, and failure-case logging. Ultrasound domain shift can arise from scanner vendor differences, acquisition protocols, and operator variability; therefore, post-deployment surveillance is essential.

X. LIMITATIONS AND THREATS TO VALIDITY

Internal validity: The present benchmark does not yet include multi-seed variance estimates or repeated cross-

```
../results/images/before_after.png
```

Fig. 3. Representative before-and-after denoising output from the project results.

```
../results/images/diff_map.png
```

Fig. 4. Difference or uncertainty-related visualization generated by the project pipeline.

validation. Observed improvements may contain run-to-run sensitivity.

Construct validity: Speckle is simulated with a multiplicative Gaussian model. Real ultrasound artifacts can deviate from this assumption, potentially changing relative performance.

External validity: Results are based on BUSI-style data and internal project artifacts. Generalization to other organs, scanners, and acquisition protocols remains to be tested.

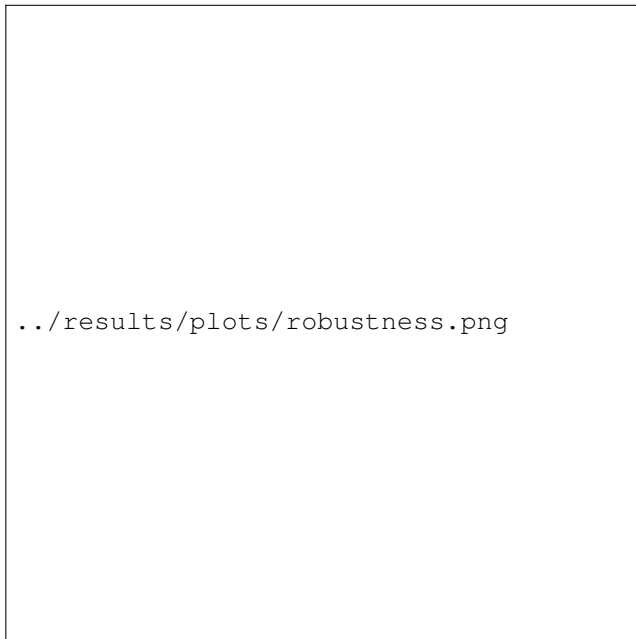


Fig. 5. Robustness curves for PSNR under different train and test noise levels.



Fig. 7. Additional qualitative preview generated from notebook-run export (20260423T133948Z).

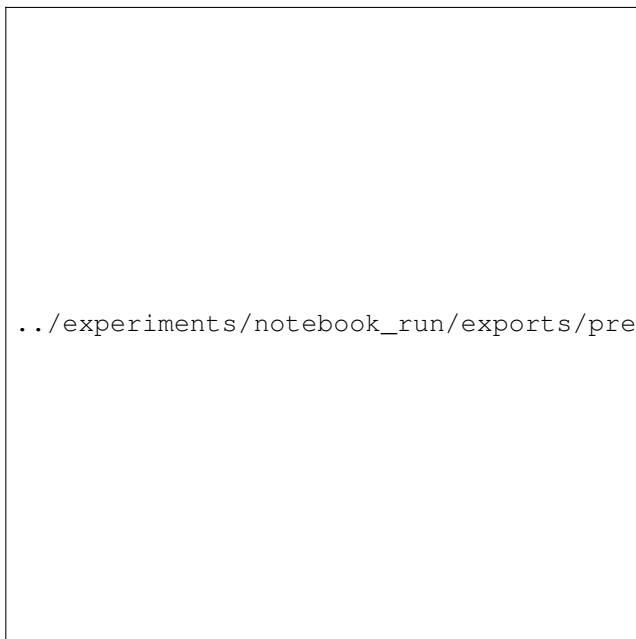


Fig. 6. Additional qualitative preview generated from notebook-run export (20260423T133845Z).

Measurement validity: MAE is available per-image in tooling but not fully exported in current aggregate tables, creating an incomplete tri-metric summary.

Comparator validity: Current quantitative reporting is strongest for project-native model variants, while some classical comparators are literature contextual rather than fully re-benchmarked under identical implementation conditions in this artifact.

Visualization validity: Uncertainty maps are qualitatively

informative, but quantitative calibration metrics (e.g., risk-coverage, expected calibration error in task-aligned formulations) are not yet reported.

XI. ETHICAL AND TRANSLATIONAL CONSIDERATIONS

Medical enhancement models should be framed as assistive tools, not autonomous diagnostic systems. Denoising may improve readability but can also alter visual perception; therefore, transparent indication of model uncertainty is important to reduce over-reliance.

For translational use, governance concerns include traceability of model versions, auditability of inference outputs, and documentation of known failure modes. Future extensions should include human-factor studies to evaluate how radiologists interpret uncertainty overlays and whether these overlays improve decision quality or only increase cognitive load.

XII. REPRODUCIBILITY AND OPEN SCIENCE STATEMENT

This paper is grounded in executable project artifacts including model code, API endpoints, and result tables. To improve reproducibility for external readers, the next release should include: (i) frozen dependency manifests, (ii) scripted experiment runners for all reported tables, (iii) seed lists and deterministic flags, and (iv) machine-readable metadata for each result file.

We emphasize that reproducibility is not binary. The present work is reproducible at prototype level and transparently identifies what is required to reach archival, publication-grade reproducibility.

XIII. CONCLUSION

This study demonstrates that uncertainty-aware U-Net denoising is a credible approach for ultrasound enhancement under speckle corruption. The enhanced model shows strong gains over baseline in available aggregate metrics, and MC dropout provides a useful spatial confidence signal that complements deterministic restoration. The work contributes both a measurable denoising improvement and a deployable analysis pipeline.

More broadly, the study argues for a dual-objective perspective in medical image restoration: optimize image fidelity while quantifying predictive reliability. This perspective is essential for trustworthy AI in clinical imaging and should become standard in future denoising benchmarks.

XIV. FUTURE WORK

Future work will focus on publication-grade validation: (1) multi-seed and fold-wise statistical testing, (2) complete MAE export in all summary tables, (3) external dataset evaluation, (4) comparison against stronger classical/deep baselines under identical protocols, and (5) uncertainty calibration studies (e.g., uncertainty-error correlation and coverage metrics).

Additional technical directions include domain-adaptive training for scanner variation, uncertainty-aware loss shaping, and integration with segmentation/classification pipelines to evaluate whether denoising-plus-uncertainty improves downstream clinical tasks.

ACKNOWLEDGMENT

The authors would like to acknowledge the ultrasound image dataset and the project implementation artifacts used to prepare this manuscript draft.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention*, 2015.
- [2] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning,” in *ICML*, 2016.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [4] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, “Image Denoising by Sparse 3-D Transform-Domain Collaborative Filtering,” *IEEE Transactions on Image Processing*, 2007.
- [5] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, “Dataset of Breast Ultrasound Images,” *Data in Brief*, 2020.
- [6] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE Transactions on Image Processing*, 2004.
- [7] N. Jayant and P. Noll, *Digital Coding of Waveforms*. Prentice Hall, 1984.
- [8] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, “Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising,” *IEEE Transactions on Image Processing*, 2017.
- [9] A. Kendall and Y. Gal, “What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?,” in *NeurIPS*, 2017.
- [10] J. Lehtinen *et al.*, “Noise2Noise: Learning Image Restoration without Clean Data,” in *ICML*, 2018.
- [11] A. Buades, B. Coll, and J.-M. Morel, “A Non-Local Algorithm for Image Denoising,” in *CVPR*, 2005.
- [12] X. Mao, C. Shen, and Y.-B. Yang, “Image Restoration Using Very Deep Convolutional Encoder-Decoder Networks with Symmetric Skip Connections,” in *NeurIPS*, 2016.
- [13] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *CVPR*, 2016.
- [14] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, “Image-to-Image Translation with Conditional Adversarial Networks,” in *CVPR*, 2017.
- [15] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely Connected Convolutional Networks,” in *CVPR*, 2017.